

The $s = 1$ edge of Hikita's $\star$: the block valuation law

$$v = \ell(\lambda) - \kappa(\lambda, \mu),$$

and two questions for you

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WIP commit: this note corresponds to commit 9791707 of `grandpa-rick/work-in-progress` (the source commit of this note; the proofs it summarizes are in `proofs/2026-10-03-day220-s1-carre-du-champ.md`, Day 220 commits `e40b350`, `db9194d`, `2324b79`; the hash line was filled in by a follow-up commit).

Status, stated once and meant throughout: the theorems below are graded *proved* in my own registry. None has been read by anyone else, and **novelty is UNAUDITED for all of them.**

1 Two questions first

Question 1 (is the $t = 0$ block law known?). Let $Q'_\lambda(x; q) = Q_\lambda[X/(1 - q); q]$ be the modified Hall–Littlewood function. Since Q'_λ and P_λ are Hall-dual and h and m are Hall-dual,

$$[h_\mu] Q'_\lambda(x; q) = \langle Q'_\lambda, m_\mu \rangle = (W(q)^{-1})_{\mu\lambda}, \quad \text{where } P_\lambda(x; q) = \sum_{\mu} W_{\lambda\mu}(q) m_\mu.$$

So $[h_\mu] Q'_\lambda$ is the coefficient of P_λ in m_μ : an entry of the *inverse* of the Hall–Littlewood $P \rightarrow m$ transition matrix. The $t = 0$ case of the block law (Theorem C below) says:

Claim ($t = 0$ block law). For $\mu \succeq \lambda$, $\mu \neq \lambda$,

$$v_{1-q}([h_\mu] Q'_\lambda(x; q)) = \ell(\lambda) - \kappa(\lambda, \mu),$$

where $\kappa(\lambda, \mu)$ is the maximal number of blocks in a simultaneous split $\lambda = \bigsqcup_i \lambda^i$, $\mu = \bigsqcup_i \mu^i$ (as multisets of parts) with $|\lambda^i| = |\mu^i|$ and $\mu^i \succeq \lambda^i$ for every i . The leading coefficient is $(-1)^{\ell - \kappa} N(\lambda, \mu)$ with $N \geq 1$ (a count of minimal raising-operator configurations, §3).

This is pure raising-operator combinatorics (proof sketch in §3; it does not use (N)), so I suspect it is folklore. **You have the Wheeler–Zinn–Justin Hall–Littlewood papers on your shelf. Have you seen this statement, or the $(1 - q)$ -adic valuation of the inverse $P \rightarrow m$ matrix, there or anywhere?** A “not found, and here is where I looked” is as useful to me as a locator.

Question 2 (making the package independent of (N)). Theorem C’s only input from (N) is the $t = 0$ edge, Day 217e Theorem B: $e_\lambda^*|_{t=0} = \omega \widetilde{H}_\lambda(x; s)$. That edge is Di Francesco–Kedem, arXiv:1505.01657, Cor. 5.18: their $M_{k,1}$ is $E_k|_{t=0}$ literally. **Proposal:** Theorem C cites DFK15 Cor. 5.18 for the $t = 0$ edge instead of (N). Then Theorems 1, A, B, C, W are all (N)-free, and the C3-nonsymmetric locator you flagged as unverified stops being load-bearing for this package. **Caveat:** I still owe you the normalization match between DFK’s conventions and ours (their (5.15), (5.25), (5.27)). I have not done it line by line. Do you agree with the citation swap, conditional on that check?

2 The results

Setting. $e_\lambda^* = E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_{\mu} c_{\lambda\mu}(s, t) e_\mu$, with the subset formula for E_k , and $v := v_{s-1}$.

The one observation. $F(X_{A^c}, sX_A) = s^{\Delta_A} F$, with Δ_A the Euler operator on the variables in A . So the $(s-1)^p$ Taylor coefficient of E_k is $\sum_A c_A X_A \binom{\Delta_A}{p}$, a differential operator of order p .

1. **Theorem 1 (biderivation).** $\partial_s(f \star g)|_{s=1} = \sum_{k,l} M_{kl} \partial_{e_k} f \partial_{e_l} g$ with $M_{kl} = \partial_s(e_k \star e_l)|_{s=1}$. It is a symmetric biderivation, i.e. the carré du champ of $L = \frac{1}{2} \sum M_{kl} \partial_k \partial_l$. At $t = 1$, (N) gives $L = \sum_i \binom{x_i \partial_i}{2}$. (Symmetry uses commutativity of $\star$.)
2. **Theorem A (lower bound).** $v(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda, \mu)$ over $\mathbb{Q}(t)$. *Proof idea:* an order- p operator glues the new part to at most p old blocks (polarization identity); then Day 214's degree count is applied block by block.
3. **Theorem C (exactness).** Equality $v(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu)$ over $\mathbb{Q}(t)$, and at $t = 0$. *Proof:* at $t = 0$, $c_{\lambda\mu}(s, 0) = [h_\mu] \tilde{H}_\lambda(x; s)$ with $\tilde{H}_\lambda(x; s) = s^{n(\lambda)} Q'_\lambda(x; 1/s)$ (Day 217e, or DFK15 per Question 2); then §3; then write $c_{\lambda\mu} = \sum_j (s-1)^j a_j(t)$, Theorem A kills a_j for $j < \ell - \kappa$, and $a_{\ell-\kappa}(0) \neq 0$. Equality holds at every t_0 outside the finite zero set of $a_{\ell-\kappa}(t)$; $t_0 = -2$ is in that set for 7 pairs at $n = 6$.
4. **Theorem B (coarsening case, (N)-free).** For μ a coarsening of λ , the leading coefficient is a sum over merge histories.
5. **Theorem W (merge weights; proved, §5b of the proof file).**

$$W_k(J) = (-1)^{|J|} [n]_t \prod_{j \in J} [k]_{t^j} / [k]_t, \quad n = k + |J|.$$

Proof from (KF) (Day 216b, (N)-free) plus Macdonald III (4.9), III.2 Ex. 1, III.7 Ex. 2. Consequence: the discrete Hall–Littlewood measure on μ_n^k integrates to the principal specialization $f(1, t, \dots, t^{k-1})$, every weight has sign $(-1)^p$ for $t > 0$, and Theorem B holds at every $t > 0$.

Correction to my own Day 219 conjecture. “ $v = \max(1, \ell(\lambda) - \ell(\mu))$ ” is **false**. First counterexample: $(2, 2, 2) \rightarrow (5, 1)$. Here $\kappa = 1$ (no block of λ has size 5 or 1), so $v = 2$, while the old formula gives 1. Computed $v = 2$ at $t = 3/5$ and $t = 7/3$.

3 Proof sketch of the $t = 0$ claim (Question 1)

Macdonald III (2.15) and the plethysm $X \mapsto X/(1-q)$ give $Q'_\lambda = \prod_{i < j \leq \ell} \frac{1-R_{ij}}{1-qR_{ij}} h_\lambda$, with $h_\alpha = 0$ if some $\alpha_i < 0$ and $h_\alpha = h_{\text{sort } \alpha}$ otherwise. Expand $\frac{1-R}{1-qR} = 1 - (1-q) \sum_{m \geq 1} q^{m-1} R^m$. A configuration is $m : \{(i, j) : i < j \leq \ell\} \rightarrow \mathbb{Z}_{\geq 0}$ with edge set $E(m) = \{m_{ij} \geq 1\}$ and result $\alpha(m) = \lambda + \sum m_{ij}(e_i - e_j)$. Then

$$[h_\mu] Q'_\lambda = \sum_{m: \alpha(m) \geq 0, \text{ sort } \alpha(m) = \mu} (-1)^{|E(m)|} (1-q)^{|E(m)|} q^{b(m)},$$

a finite sum. Every configuration contributes exactly $(-1)^{|E|}$ at order $(1-q)^{|E|}$, so **there is no cancellation at lowest order:** $v_{1-q} = E_{\min}$ and the leading coefficient is $(-1)^{E_{\min}} N$.

- $E_{\min} \geq \ell - \kappa$: transfers stay inside connected components of $([\ell], E(m))$ and go from larger to smaller index, so each component C satisfies $\text{sort}(\alpha|_C) \supseteq \lambda_C$. The components therefore form a compatible block split, so $\#\text{components} \leq \kappa$ and $|E| \geq \ell - \kappa$.
- $E_{\min} \leq \ell - \kappa$: in a κ -block split every block (ρ, ν) is indecomposable, which forces strict partial-sum inequalities $B_r > P_r$ for $r < L$. The path configuration $m_{c_r, c_{r+1}} = B_r - P_r$ realises ν with $L - 1$ edges per block.

Independent check (`raising_t0.py`, a different engine from the subset formula): on all 84 pairs with $n = 4-6$, the $t = 0$ valuation and the leading coefficient match.

4 Evidence and honest grades

- Theorems 1, A, B, C, W: **proved** (my registry; not peer-read). Novelty: **UNAUDITED** for all of them.
- Theorem C's input from (N) is the $t = 0$ edge only. If (N) falls and the DFK15 swap is not accepted, Theorem C falls back to “proved on coarsenings (Theorem B) + computed elsewhere”.
- Computed, generic t : $n \leq 5$ symbolic t , 35/35; $n = 6$, 53/53 at $t = 3/5$ and 53/53 at $t = 7/3$; $n = 7$ at $t = 3/5$, 87/87 pairs logged (**partial run**: the process stopped before finishing).
- $t = 0$: 84/84 pairs over $n = 4$ –6, by two engines.
- Theorem W: 112/112 through $n = 8$; merge-history formula reproduces 132/132 logged coarsening leading coefficients at $n = 6$, $t \in \{3/5, 7/3, -2\}$, including the zeros at $t = -2$.
- Failures at $t = -2$: 7 pairs at $n = 6$ with $v > \ell - \kappa$, all explained by $a_{\ell-\kappa}(-2) = 0$.

What I would like from you. Questions 1 and 2 above, in that order. Grading the theorems can wait for your queue; I know the (s, t) -square is already three deferrals deep, and I am not asking you to jump this ahead of it.